

Emergent Structures as Factorization of an Anonymous Complex Wave Multiset: Working Hypotheses and a Numerical Verification Program for Simplices, Localization, Condensates, and Dynamics

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Abstract

This paper formulates a working hypothesis that asks whether structures such as space, time, particles, vertices, edges, and complete networks can be uniquely read out from a multiset of anonymous complex waves alone, without being assumed as foundational.

Conventionally, when describing relations among multiple elements, one first fixes the number of elements and counts all pairwise relations as the number of edges of a complete graph. This description, however, implicitly introduces identifiable and persistent vertices, endpoints for each relation, complete connectivity, and a one-relation-one-edge correspondence. If the underlying waves are anonymous, these cannot be axioms; they must be reconstruction results that are admitted only after a successful factorization of the anonymous set.

We define the base state as a complex wave multiset identified under permutations. A submultiset is read as a harmonic series, a localization, a condensate, a simplex, or a persistent object only when it is uniquely factorized while simultaneously satisfying conditions such as zero-square closure, finite recurrence, integer ratios, common phase, low rank, and complex Gram consistency. The complete network is not an input; it is a connectivity structure that becomes known as a consequence, once anonymous relational quantities have been uniquely reconstructed into a simplex.

Furthermore, we define dynamics not as trajectories of named constituents, but as a process in which the factorization type of the anonymous wave set is preserved while its common phase center changes. We propose ground-truth embedding experiments in which all provenance information of data generated from known structures is completely erased, blind control experiments with no planted structure, permutation-invariance tests, perturbation tests, and temporal persistence tests. This paper does not prove the existence or uniqueness of emergent structures. It fixes a falsifiable problem setting for verifying them without breaking anonymity.

Keywords: anonymous complex waves, factorization, zero-square closure, finite recurrence, simplex, unlabeled distance geometry, localization, condensate, emergent dynamics, numerical experiments

1. Introduction

When constructing a physical model, it is natural to number particles or nodes and to describe the relations among them as arrays or matrices. If the number of elements is N and one relation is placed between every pair of elements, the number of relations is

$$M = \frac{N(N-1)}{2}.$$

This counting itself is correct. Before it, however, at least the following assumptions have been made.

1. N nodes exist in advance.
2. The nodes can be distinguished from one another.

3. The identity of each node is preserved throughout any change.
4. A relation exists between every pair of nodes.
5. Each relation has one pair of endpoints.
6. Relations among three or more elements are composed of pairwise relations.

If, however, the only underlying entities are anonymous complex waves, then the waves carry no vertex numbers and no edge numbers from the start. Array indices are storage locations in a computation, not physical names. Nevertheless, if relations are represented by index pairs from the outset, the complete network and its connectivity information are smuggled into the foundation.

This paper reverses that order. Instead of starting from

complete network $\longrightarrow$ relational quantities,

we ask about

anonymous complex wave set $\longrightarrow$ unique factorization $\longrightarrow$ readout of geometry, localization, condensates, and dynamics

With this change, the complete network ceases to be an ontological premise. Only when a part of the anonymous relational quantities is uniquely reconstructed into a simplex with some number of vertices do we judge that this subset could be read as the edge set of a complete graph.

The same idea applies to harmonics. We do not first posit a base wave and its child harmonics. When a subset of the anonymous wave set is found that can be factorized with a common fundamental period and integer ratios, we read it, as a result, as having been a harmonic series. Localization, condensation, parent-child relations, and motion are likewise defined as stable factorization results of the anonymous set.

2. Position and Scope of Claims

This paper is neither a proof paper nor a report of numerical results. It has three aims.

1. Define a base state that preserves anonymity.
2. Define emergent structures as factorizations of the anonymous set.
3. Fix experimental procedures that verify the hypotheses without circularity and can detect failure.

We do not claim that space, time, or particles have already been derived. We do not claim that any particular update rule is the correct physical law. Nor do we claim any correspondence with real particles, the Standard Model, gravity, or cosmology.

The central proposition proposed in this paper is the following.

If a small number of strongly consistent factors exist within an anonymous complex wave multiset, they may be uniquely read out, independently of labels, as simplices, harmonics, localizations, condensates, or persistent objects.

3. Anonymous Complex Wave States

3.1 Base set

Computationally we handle M complex waves arranged in order:

$$W = (w_1, w_2, \dots, w_M), \quad w_m \in \mathbb{C}.$$

The indices, however, are not physical names. We regard any permutation as the same state and define the base state as the following equivalence class:

$$\mathcal{W} = [W]_{S_M} = \{(w_{\pi(1)}, \dots, w_{\pi(M)}) \mid \pi \in S_M\}.$$

The physical input is therefore not an ordered vector but a multiset:

$$\mathcal{W} = \{w_1, \dots, w_M\} / S_M.$$

Since several waves may share the same value, we treat this as a multiset rather than an ordinary set.

3.2 Zero-square closure

The base constraint adopted in this paper is the closure of the complex sum of squares, not of the Hermitian norm:

$$Q(\mathcal{W}) = \sum_{m=1}^M w_m^2 = 0.$$

This expression does not depend on the ordering of the waves. It is also closed under a common rescaling by any nonzero complex number:

$$Q(\lambda \mathcal{W}) = \lambda^2 Q(\mathcal{W}) = 0.$$

Hence the absolute scale is not fixed by this condition alone.

3.3 Finite recurrence

We define finite recurrence not as the condition that each named wave returns to the same place, but as the condition that the anonymous set as a whole returns to the original state up to permutation equivalence:

$$U^q \mathcal{W} \simeq \mathcal{W}.$$

Here q is the finite recurrence order, and the equivalence includes at least permutations of the waves. When necessary, common phase, overall scale, or geometric congruence transformations are also included in the equivalence relation.

Under this definition, each array element need not return to the same index after one period. It suffices that the same state is reconstituted as a set.

3.4 Distinguishing computational order from emergent time

Numerical experiments require step numbers for performing updates:

$$s = 0, 1, 2, \dots$$

This s , however, is the order in which the computer processed the updates, not physical time. We call it the computational order. If, in the future, a factor readable as time is constructed, it must be reconstructed separately from finite cycles or reference phases within the anonymous wave set:

$$\tau_{\text{read}} = F_{\text{clock}}(\mathcal{W}_s, \mathcal{W}_{s+1}, \dots).$$

4. Anonymous Factorization

4.1 Definition of a factor

We express the operation of reading structure from an anonymous wave state as a factorization map:

$$\mathcal{F} : \mathcal{W} \mapsto \{F_1, F_2, \dots, F_L\}.$$

A factor F consists of a submultiset, latent parameters, a reconstruction rule, and a residual:

$$F = (C, \theta, R, \varepsilon), \quad C \subseteq \mathcal{W}.$$

Here C is the anonymous submultiset constituting the factor; θ denotes latent parameters such as a common period, a phase center, integer orders, or a geometric dimension; R is the rule that reconstructs C from them; and ε is the reconstruction residual.

For an exact factor,

$$\varepsilon(F) = 0.$$

In numerical experiments, allowing for finite precision, a factor is called approximate when its residual is below a tolerance fixed in advance:

$$\varepsilon(F) \leq \varepsilon_{\text{tol}}.$$

4.2 Equivalent factorizations

If two factorizations differ only by permutations of the input waves, permutations of factor names, a global phase, a common scale, a mirror reflection, or a congruence transformation of the adopted geometry, we regard them as the same physical interpretation.

The number of inequivalent valid factorizations is denoted by

$$\Omega_{\text{fact}}(\mathcal{W}) = \# \left(\frac{\{\text{valid factorizations}\}}{\text{admissible symmetries}} \right).$$

The judgment falls into three cases:

$$\Omega_{\text{fact}} = 0 \implies \text{not reconstructible as the specified structure,}$$

$$\Omega_{\text{fact}} = 1 \implies \text{uniquely reconstructed up to physical equivalence,}$$

$$\Omega_{\text{fact}} > 1 \implies \text{multiple inequivalent structural interpretations exist.}$$

Finding one planted ground truth demonstrates only existence. To claim unique emergence, one must confirm that no inequivalent alternative solutions exist.

5. Simplex Reconstruction without Inputting a Complete Network

5.1 Anonymous set of complex squared lengths

Take a submultiset as candidate complex squared lengths:

$$D = \{d_1^2, d_2^2, \dots, d_r^2\}.$$

At this stage, no vertices and no edges exist, and no correspondence between lengths and point pairs is given.

If a candidate vertex number n exists, the required number of lengths satisfies

$$r = \frac{n(n-1)}{2}.$$

Here this equation is not a foundational axiom. It is a necessary condition for judging whether the anonymous subset can serve as candidates for all edges of an n -vertex simplex.

5.2 Provisional edge assignment

The reconstructor examines all essentially distinct assignments from the anonymous length set to candidate edges:

$$f : D \longrightarrow \{\{i, j\} \mid 0 \leq i < j < n\}.$$

This map is not base data; it is a candidate variable of the inverse problem.

Taking vertex 0 as the reference, a complex Gram matrix is constructed from the assignment f :

$$G_{ab} = \frac{1}{2} (d_{0a}^2 + d_{0b}^2 - d_{ab}^2), \quad 1 \leq a, b < n.$$

Only assignments satisfying the required embedding dimension, rank, nondegeneracy, closure conditions, and any additional symmetry conditions are accepted as valid. Because complex squared lengths are involved, positivity and triangle inequalities valid for real Euclidean distances are not required unconditionally. Which congruence transformations are identified is specified in the experimental protocol together with the adopted complex bilinear form.

5.3 The complete network is a reconstruction certificate

Only when exactly one valid assignment remains, up to vertex permutations and congruence transformations, do we judge that D uniquely determines an n -vertex simplex:

$$D \xrightarrow{\text{unique reconstruction}} \text{n-vertex simplex.}$$

Only then does it become known that the elements of D corresponded to all edges of a complete graph:

$$K_n \text{ is a reconstruction result, not an input.}$$

The problem of reconstructing real point configurations from unlabeled distance multisets has been studied as unlabeled distance geometry and unlabeled global rigidity. This paper shares that problem awareness, but adds, as working hypotheses, that the input is not restricted to positive real distances and that complex waves, zero-square closure, finite recurrence, harmonic factors, and factor persistence are treated within a single readout framework.

6. Factorization of Harmonics, Localization, and Condensation

6.1 Harmonics are not an a priori parent-child relation

We do not posit a base wave first and generate integer multiples as child waves. From the phase histories or cyclic properties of the anonymous wave set, we search for a subset C for which a common fundamental quantity and an integer sequence exist:

$$\omega_m \simeq k_m \omega_0, \quad k_m \in \mathbb{Z}, \quad w_m \in C.$$

A wave corresponding to the fundamental quantity need not be explicitly present in the set. The fundamental can be inferred inversely as a latent factor shared by several waves.

For phases too, we search for the following low-dimensional representation:

$$\phi_m(s) \simeq k_m \phi_0(s) + \delta_{k_m} \pmod{2\pi}.$$

If this representation holds with a small residual, C is read as a harmonic factor.

6.2 Localization is a readout of common phase

The readout kernel obtained by synthesizing a harmonic factor is written as

$$L_C(\phi) = \left| \sum_{m \in C} A_m \exp [ik_m(\phi - \phi_0) + i\delta_{k_m}] \right|^2.$$

When L has a sharp peak near the phase ϕ_0 , the factor is read as a localized state. This does not mean that the underlying waves exist at a point in a background space. It means that the common phase factor of the anonymous wave set has a sharp coincidence position with respect to a particular readout.

If the highest harmonic order is K , then for an ideal series with equal amplitudes and regular phases, the phase resolution improves roughly at the order

$$\Delta\phi \sim \frac{1}{K}.$$

Even when many harmonics are present, if they are all generated from one common fundamental phase and a fixed integer sequence, the number of independent degrees of freedom need not have grown with the number of harmonics. The very fact that many components can be described by a few latent factors serves as an indicator of condensation or low-rankness.

6.3 Condensates

In this paper a condensate is defined not as a fixed set of named elements but as a persistent low-dimensional factor satisfying the following.

1. Many anonymous waves can be reconstructed from a few latent parameters.
2. The partial zero-square closure, or a specified conservation residual, is small.
3. It has a common phase center and a finite phase width.
4. Its factor signature, modulo admissible symmetries, is preserved across computational order.

The constituent waves of a condensate may be exchanged during updates. What is conserved is not the names of elements but the factorization type.

7. Symmetry and Uniqueness

7.1 The need for strong symmetry

Sets that simultaneously satisfy zero-square closure, finite recurrence, integer ratios, low rank, and geometric consistency are expected to be special among generic random complex wave sets. Strong symmetry is a candidate condition making it possible to compress many anonymous components into a few factors.

On the other hand, perfect symmetry produces a degeneracy in which no vertex, direction, or phase center can be distinguished. What is needed is therefore not simply maximal symmetry. We take as a working hypothesis the two-layer structure

strong symmetry enabling factorization + a small symmetry breaking selecting the readout.

7.2 Identification of symmetric solutions

The many solutions obtained by renaming the vertices of a regular simplex are not different physical structures. Uniqueness is therefore judged not by the number of labeled solutions but by the number of equivalence classes after quotienting by admissible symmetries.

If, however, distinct simplices, distinct harmonic fundamentals, or distinct condensate partitions remain that cannot be connected by symmetry transformations, that is genuine non-uniqueness.

7.3 Perturbation regimes

Starting from a perfectly symmetric state, we add perturbations that preserve anonymity:

$$w_m(\epsilon) = w_m^{(0)} + \epsilon \eta_m.$$

We test for the possible existence of three regimes.

1. Perfectly symmetric point: the shape is clear but directions or centers are degenerate.
2. Small-perturbation regime: the shape is preserved while a position or direction is selected.
3. Large-perturbation regime: valid factors are lost and reconstruction becomes non-unique or impossible.

The existence of a critical perturbation magnitude is itself a working hypothesis and may be refuted by experiment.

8. Dynamics as Preservation of Factorization Type

8.1 Microscopic updates and observed dynamics

A microscopic update is expressed as a map producing a sequence of anonymous states:

$$\mathcal{W}_{s+1} = \Phi(\mathcal{W}_s).$$

The update map must not depend on input ordering:

$$\Phi(\pi\mathcal{W}) \simeq \Phi(\mathcal{W}).$$

At each computational order we perform a factorization:

$$\mathcal{F}(\mathcal{W}_s) = \{F_1(s), F_2(s), \dots\}.$$

Observed dynamics is defined not as trajectories of individual waves but as changes of factor equivalence classes:

$$[F_\alpha(s)] \longrightarrow [F_\beta(s+1)].$$

8.2 Factor signature

To judge the identity of a factor, we define a signature that does not depend on element names:

$$\Sigma(F) = (q, \{k_m\}, \text{amplitude ratios, closure residual, Gram spectrum, rank, localization width}, \dots).$$

The concrete signature components are preregistered for each target factor. When the signature agrees before and after an update and only the common phase center changes, the factor is read as a conserved condensate.

8.3 Motion

With factor shape C_0 and common phase translation T , a moving factor is expressed as

$$F(s) \simeq T_{\phi(s)} F^{(0)}.$$

If after the update

$$\Sigma(F(s+1)) \simeq \Sigma(F(s))$$

still holds, the lump is conserved. Meanwhile the center phase may change:

$$\phi(s+1) \neq \phi(s).$$

Motion is therefore not the same named material changing its position; it is the persistence of the same factorization type while its common phase center changes.

8.4 Fusion, splitting, and interactions

Recompositions of the factorization are read as follows:

$$\{F_A, F_B\} \longrightarrow \{F_C\} \quad \text{fusion candidate,}$$

$$\{F_A\} \longrightarrow \{F_B, F_C\} \quad \text{splitting candidate,}$$

$$\{F_A, F_B\} \longrightarrow \{F'_A, F'_B\} \quad \text{scattering or exchange candidate.}$$

To call these physical interactions, one must identify global invariants that are conserved independently of changes in the number of factors. This paper does not assume their concrete form.

9. Working Hypotheses

We divide the claims of this paper into forms that future experiments can refute individually.

Hypothesis H1: Anonymous simplex reconstruction

Some anonymous sets of complex squared quantities with strong consistency are uniquely reconstructed into a single complex simplex, up to vertex permutations, congruence transformations, and admissible gauges.

Hypothesis H2: A posteriori appearance of the complete network

For a subset admitting a unique simplex reconstruction, the complete graph structure becomes known as a result, from the number of elements and the connection assignment. There is no need to input a complete network.

Hypothesis H3: Localization readout by harmonic factors

An anonymous wave subset sharing integer ratios, a common fundamental phase, and regular amplitudes produces a phase peak of finite width in the synthesized readout. The center and width of this peak are read as the localization position and localization width.

Hypothesis H4: Condensates as low-rank factors

When many anonymous waves are reconstructible from a few latent factors and the closure and phase width remain stable, the factor persists as a condensate.

Hypothesis H5: Motion as preservation of factor type

When the signature of a condensed factor is preserved and only its common phase center changes continuously, the change is read as motion.

Hypothesis H6: Emergent space, time, and objects

When a simplex factor, an internal cyclic factor, and a persistent condensed factor are independently reconstructed from the same anonymous wave sequence and are mutually consistent, they can be read respectively as space, time, and objects.

Hypothesis H7: Symmetry with small breaking

There exists a finite regime in which strong symmetry enabling unique factorization coexists with a small symmetry breaking that selects positions and directions.

10. Numerical Verification Program

10.1 Complete separation of generator and reconstructor

The experiments consist of two independent programs.

The generator produces complex wave sets from known structures:

$$S_{\text{true}} \xrightarrow{\text{generator}} \mathcal{W}.$$

After generation, all of the following are erased:

- vertex numbers
- edge endpoints
- generation order of waves
- harmonic orders
- fundamental-wave labels
- condensate labels
- parent-child labels
- individual IDs across computational order
- coordinates used in generation

The input ordering is further randomized.

The reconstructor receives only the permitted complex values, phase histories, and global constraints. The reconstructor must not access the generator's internal data, ground-truth labels, or connection tables.

10.2 Experiment E0: Permutation-invariance test

The same anonymous set is input under multiple random permutations:

$$\mathcal{W}^{(r)} = \{w_{\pi_r(1)}, \dots, w_{\pi_r(M)}\}.$$

We require that all inputs yield the same factorization result up to admissible symmetries. If inequivalent results depend on the input ordering, the implementation has a hidden label dependence.

10.3 Experiment E1: Ground-truth embedded simplex

All pairwise relational quantities are generated from a known real or complex simplex, and the correspondence table is erased. We test whether the reconstructor satisfies the following.

1. It recovers a solution congruent to the original simplex.
2. No inequivalent alternative solution exists.
3. Even without inputting the vertex number, it recovers the candidate vertex number.
4. The complete network is obtained as an output result.

10.4 Experiment E2: Ground-truth embedded harmonics and localization

Wave sets are generated from a known fundamental phase and integer orders and then anonymized. We test whether the reconstructor can recover the fundamental quantity, the set of integer orders, the phase center, and the localization width.

We also test the case where the wave corresponding to the fundamental is removed from the input set, checking whether the latent fundamental factor can be inferred inversely.

10.5 Experiment E3: Composite factors

A simplex factor, a harmonic factor, and a closure factor are simultaneously embedded in the same anonymous wave set. We verify whether the individually recovered factors occupy the same subsets or consistent overlaps.

This experiment examines whether different readouts point to the same emergent structure.

10.6 Experiment E4: Persistence and motion lock

We generate sequences of anonymous sets in which the factor shape is preserved and only the common phase center is varied. At each computational order, the wave ordering and the constituents are shuffled, making named tracking impossible.

We verify whether the reconstructor can recover the same factor signature and the change of phase center without using the identity of the constituent waves.

10.7 Experiment E5: Blind controls

Without planting any structure, we generate anonymous sets satisfying only zero-square closure, finite recurrence, or the same marginal distributions. Compared with the ground-truth embedded group, we measure the frequencies of valid factors, unique factors, and false positives:

$$P_{\text{unique}} = P(\Omega_{\text{fact}} = 1 \mid \text{generating condition}).$$

If structures are detected at the same frequency in the control group as in the ground-truth group, the reconstruction conditions are too weak.

10.8 Experiment E6: Perturbation and criticality

Controlled perturbations are added to ground-truth embedded sets, and the following are measured:

- number of factorizations
- ground-truth recovery rate
- reconstruction residual
- Gram rank and spectrum
- localization peak width
- lifetime of factor signatures
- false-positive rate

The regime from the perfectly symmetric point to factor collapse is scanned continuously.

11. Success Conditions

The minimal success conditions of this research program are set as follows.

1. Changing the input ordering yields the same physical equivalence class.
2. The ground-truth structure can be recovered without using names, coordinates, or connectivity information from generation.
3. No alternative solution exists that is not symmetry-equivalent to the ground truth.
4. Multiple independent readouts point to the same factor.
5. Factors have a finite stability regime against small perturbations.

6. Persistent objects can be identified solely by conservation of factor signatures, without tracking constituent waves.
7. The computational step itself is not taken as physical time; a time candidate can be reconstructed from internal cycles.
8. Verifiable predictions about the next factor state can be obtained from the current factor state.

Constructing separate readouts for space, time, and objects is not sufficient. We require that they be mutually consistent on the same anonymous wave set and constrain each other's predictions.

12. Refutation Conditions

If any of the following holds, the corresponding hypothesis is not supported.

1. Inequivalent structures emerge depending on the ordering of the input array.
2. Even when the ground truth is recovered, many equally valid inequivalent solutions remain.
3. Structures can be recovered only if ground-truth labels or connection tables are passed to the reconstructor.
4. Structures are detected in random control groups at rates comparable to ground-truth embedded groups.
5. Factors are completely destroyed by arbitrarily small perturbations.
6. The localization readout depends on external coordinates used at generation time.
7. The judgment of motion depends on tracking named constituent waves.
8. Emergent time is nothing but an alias of the computational step.
9. The simplex readout, the harmonic localization readout, and the persistent factor readout contradict one another.
10. Expanding the search range destroys uniqueness, revealing that the earlier unique solution was an artifact of insufficient search.

Failure results are also important. The conditions under which factorization becomes impossible, non-unique, or unstable delimit the regime in which emergent structures hold.

13. Computational Cost and Search Strategy

A naive search assigning r anonymous candidate lengths to the edges of an n -vertex simplex grows rapidly. In the generic case where all values are distinct, an estimate of the number of candidates up to vertex permutations is

$$\frac{r!}{n!}, \quad r = \frac{n(n-1)}{2}.$$

For large systems, therefore, instead of exhaustive permutation search, the following pruning becomes necessary:

- partial Gram consistency
- rank conditions
- subsets with zero-square closure
- finite cycle orders
- integer-ratio candidates
- repeated values and automorphisms
- spectral invariants
- low-rank decompositions
- gluing consistency of partial simplices

In ground-truth embedding experiments the generator knows the answer, but the reconstructor must not use that answer for pruning. The ground truth is used only for evaluation after the reconstruction has finished.

When the search becomes large-scale, constructing uniqueness certificates instead of enumerating all solutions will also be considered. However, uniqueness must not be claimed merely because one locally optimal solution has been found.

14. Minimal Relation to Existing Work

The question of whether real point configurations can be reconstructed from unlabeled distance multisets has been studied. Boutin and Kemper treated the problem of reconstructing point configurations from the distribution of distances or areas, showing that many generic configurations are reconstructible while counterexamples also exist. Gortler, Theran, and Thurston studied the relation between unlabeled distances and global rigidity in generic position.

This paper does not claim that these results apply as they stand to complex wave systems. Handling complex squared quantities lacking the positivity of real positive distances, zero-square closure, finite recurrence, harmonic factors, and factor persistence across time simultaneously requires a different formulation and numerical verification.

15. Limitations and Future Work

At the stage of this paper, the following remain unresolved.

1. Whether anonymity should quotient only by permutations, or whether wider basis transformations should also be identified.
2. The congruence group of complex squared lengths and the exact equivalence relation for unique reconstruction.
3. Whether the global structure in the case of overlapping factors becomes a partition, a covering, or a hypergraph.
4. Natural measures and tolerances for valid factor residuals.
5. Conditions under which strong symmetry increases unique factorizations, versus conditions under which it increases degeneracy.
6. Concrete procedures for constructing emergent time from internal cycles.
7. Microscopic update rules that preserve anonymity while producing interactions between factors.
8. Global invariants conserved even when the number of factors changes.
9. Uniqueness-certifying algorithms enabling large-scale search.

These are not concealments of gaps in this paper; they are explicit tasks connecting the working hypotheses to the next experiments.

16. Conclusion

This paper proposed a research framework that does not place complete networks, harmonic parent-child relations, localized bodies, or particles at the foundation, but reads them out solely from an anonymous complex wave multiset.

Taking

$$\mathcal{W} = \{w_1, \dots, w_M\}/S_M$$

as the base state, valid partial factors are searched for:

$$\mathcal{W} \xrightarrow{\text{anonymous factorization}} \left\{ \begin{array}{l} \text{simplex factors} \\ \text{harmonic factors} \\ \text{localization factors} \\ \text{condensate factors} \\ \text{persistent factors} \end{array} \right\}.$$

When a simplex factor is uniquely reconstructed, the complete network turns out to be its reconstruction result, not an input. Harmonic relations likewise appear, without positing a parent-child structure first, as factorization results of subsets sharing a common fundamental quantity and integer ratios. Localization is a sharp readout of a common phase factor, and a condensate is a low-rank factor describing many anonymous waves by a few latent quantities.

Dynamics is defined not as trajectories of named waves but as changes of factor equivalence classes that preserve the factor signature while changing the common phase center:

$$[F(s)] \longrightarrow [F(s + 1)].$$

The success or failure of this framework can be judged by blind reconstruction of anonymized ground-truth data, control experiments with no planted structure, permutation-invariance tests, perturbation tests, and persistence tests.

The central proposal of this paper is not to assume structures and assign them to waves. It is to ask

under which conditions an anonymous wave set is uniquely factorized as a single structure.

If factors readable as space, time, and particles are reconstructed consistently with one another from the same anonymous wave set, without using names or coordinates from generation time, the minimal goal of this research program is achieved.

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Appendix A: List of Symbols

Symbol	Meaning
M	total number of anonymous complex waves in the base input
W	ordered complex wave array in computation
$\mathcal{W}$	anonymous complex wave multiset identified under permutations
Q	closure quantity given by the complex sum of squares
q	finite recurrence order
s	update order in numerical computation; not physical time
F	factor extracted from the anonymous set
C	submultiset constituting a factor
θ	latent parameters of a factor
ε	factor reconstruction residual
Ω_{fact}	number of inequivalent factorizations modulo admissible symmetries
D	anonymous set of complex squared quantities used for simplex reconstruction

Symbol	Meaning
n	candidate number of simplex vertices for reconstruction
G	complex Gram matrix built from a candidate assignment
k	integer order of a harmonic factor
ϕ_0	recovered common phase center
Σ	invariant signature of a factor

Appendix B: Items to Preregister for Experiments

Before running each numerical experiment, at least the following are fixed.

1. Data items permitted as input
2. Generation information hidden from the reconstructor
3. Symmetry transformations to be identified
4. Definition of factor residuals
5. Numerical tolerances
6. Search ranges and cutoff conditions
7. Method for judging uniqueness
8. Method for evaluating ground-truth recovery
9. Method for generating random control groups
10. Perturbation distributions
11. Success conditions
12. Refutation conditions

This preregistration prevents the circularity of conveniently changing factor definitions, tolerances, or search ranges after seeing the results.